Ahmed Essam Ahmed

Machine Learning Engineer

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PROFILE

Computer Science and Mathematics student at Helwan University, specializing in Data Science and End-to-End Machine Learning systems. Combining a strong foundational background in statistics and predictive modeling with hands-on technical expertise in Python, SQL, and data visualization tools. Experienced in building high-accuracy predictive systems, conducting advanced EDA, and translating complex datasets into actionable business insights.

EDUCATION

Bachelor's degree, Faculty of Science, Computer Science & Mathematics Department 2023 – 2027
Helwan University

SKILLS

Programming & Tools
Python, SQL, Git & GitHub, Jupyter Notebook, VS Code , Google Colab

Machine Learning
Scikit-learn, Supervised Learning, Unsupervised Learning, Boosting Algorithms, Hyperparameter Tuning, Model Evaluation, Feature Engineering

Communication & Team Collaboration

Adaptability & Continuous Learning

Data Analysis & Visualization
Pandas, NumPy, Matplotlib, Seaborn Power BI, EDA

Problem-Solving & Analytical Thinking

Time Management

PROFESSIONAL EXPERIENCE

AI & Data Science Trainee 01/2026 – Present
EM Business Solutions Cairo, Egypt (Remote)

- **Machine & Deep Learning:** Built and optimized regression, classification, and clustering models using Scikit-learn (ensemble methods & hyperparameter tuning); developed Deep Learning and Computer Vision solutions using CNNs, transfer learning, and YOLO.
- **NLP & Large Language Models:** Implemented NLP pipelines (text preprocessing, TF-IDF, Word2Vec, GloVe) and worked with transformer-based BERT/GPT architectures within the Hugging Face ecosystem.
- **Generative AI & Chatbot Deployment:** Built GenAI applications using prompt engineering and RAG with ChromaDB and LangChain, designing and deploying an end-to-end domain-specific AI chatbot.
- **Deployment & Containerization:** Deployed production-ready AI applications using Streamlit and FastAPI, utilizing containerized workflows with Docker.

Machine Learning Engineer Trainee 11/2025 – Present
Digital Egypt Pioneers Initiative (DEPI) Cairo, Egypt (Hybrid)

- **End-to-End ML Pipelines:** Developed and optimized end-to-end Machine Learning pipelines, managing the full lifecycle from data preprocessing and feature engineering (Pandas, NumPy, Scikit-learn) to model deployment.
- **Advanced AI Architectures:** Built and evaluated Deep Learning architectures, implementing Natural Language Processing (NLP) for text analysis and Computer Vision (CV) models for image-based tasks.
- **MLOps & Lifecycle Management:** Gained hands-on experience in MLOps practices, utilizing MLflow for model tracking, versioning, and managing lifecycle stages.

- **Cloud Integration:** Scaled machine learning workflows and automated model training by integrating training pipelines with Microsoft Azure AI services.

Machine Learning Engineering Intern — 09/2025 – 10/2025
Elevvo Pathways — Cairo, Egypt (Remote)

- **End-to-End Pipelines:** Built and optimized modular, end-to-end Machine Learning pipelines for regression, classification, and clustering tasks using Python.
- **Feature Engineering:** Conducted robust data preprocessing, advanced feature engineering, and rigorous model evaluation utilizing **Pandas, NumPy, and Scikit-learn**.
- **Workflow Optimization:** Developed and structured systematic model training, hyperparameter tuning, and testing workflows to improve baseline model performance and interpretability.

Advanced Data Analytics Intern — 07/2025 – 08/2025
National Telecommunication Institute (NTI) — Cairo, Egypt (Remote)

- **Data Engineering & EDA:** Executed data cleaning, feature extraction, and exploratory data analysis (EDA) on diverse real-world datasets using Python.
- **Business Intelligence:** Designed and developed interactive dashboards and data products using **Power BI** to translate complex analytical workflows into actionable business insights.
- **Predictive Modeling:** Built and evaluated core machine learning models (classification & regression), collaborating with cross-functional peers to optimize analytical workflows.

PROJECTS

Star Classification Prediction (LightGBM)

- Data Pipeline: Cleaned and engineered features from 100,000+ astronomical observations (SDSS dataset) using Python.
- Model Development: Benchmarked multiple algorithms (XGBoost, Random Forest, SVM) and selected LightGBM for optimal performance.
- Key Outcome: Achieved ~98% predictive accuracy and leveraged SHAP for model interpretability.

Student Score Prediction
Tools: Python (Scikit-learn , Matplotlib , Pandas)

- Collected and preprocessed a dataset containing students' study hours and participation data.
- Performed data cleaning, exploratory data analysis (EDA), and visualization to understand performance patterns.
- Built and trained regression models to predict students' exam scores based on study behavior.
- Evaluated model performance and identified key factors influencing student success.

Customer Segmentation
Tools: Python (Scikit-learn , Matplotlib , Pandas , seaborn)

- Collected and prepared a mall customer dataset containing income and spending score attributes.
- Applied K-Means clustering to segment customers into distinct groups based on purchasing behavior.
- Performed data visualization and cluster analysis to identify patterns and customer profiles.
- Extracted actionable marketing insights to enhance customer targeting and business strategy.

CERTIFICATES

Advanced Data Analytics – NTI (Summer Training)
Jul 2025 – Aug 2025

Machine Learning Engineering Internship – Elevvo Pathways
Sep 2025 – Oct 2025

AI and Machine Learning Foundations – Sprints
2025

LANGUAGES

Arabic
Mother Tongue

English
Very Good